



S&T **Git &** **Github**



Introduction

Why do we learn git?
Why is this the first session?

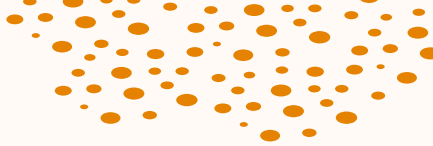


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01

Git

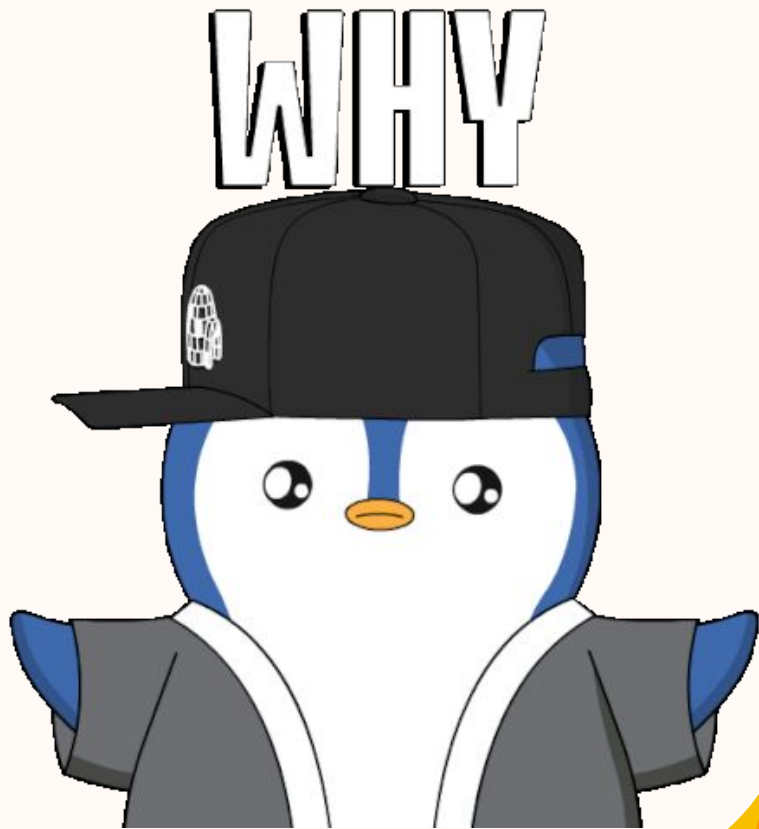
Why, What, How



Why do we need Git?

- To manage versions of code efficiently.
- lets multiple developers collaborate.

Well, it's simply to prevent “**final_final_v3.cpp**” disasters 😅



What is Git?

Git is a Distributed Version Control System
DVCS

What does Git offer?

- It keep track of all changes made to files.
- Provides a complete history of a project.
- Allow you to rollback into an older version.



How to use Git?

Git offers us a working directory(**Folder**) to view our project and update it easily.

There are 3 main stages:

- Working directory
- Staging area
- Git directory





Who are you?

Set username

```
git config --global  
user.name "Ibrahim"
```

Set email

```
git config --global  
user.email  
"iymetwally@gmail.com"
```



How to save your work



git init

Used to initialize a Git repository in the current directory.



git add

Used to add some files into the Staging area.



git commit

Used to submit all files on the Staging area into the Git repository.

What did I change?



To know that you can use:

git status

Used to display information about each file if it is modified or not or at the staging area,

git log

Used to display commit history.





Hands-on 1

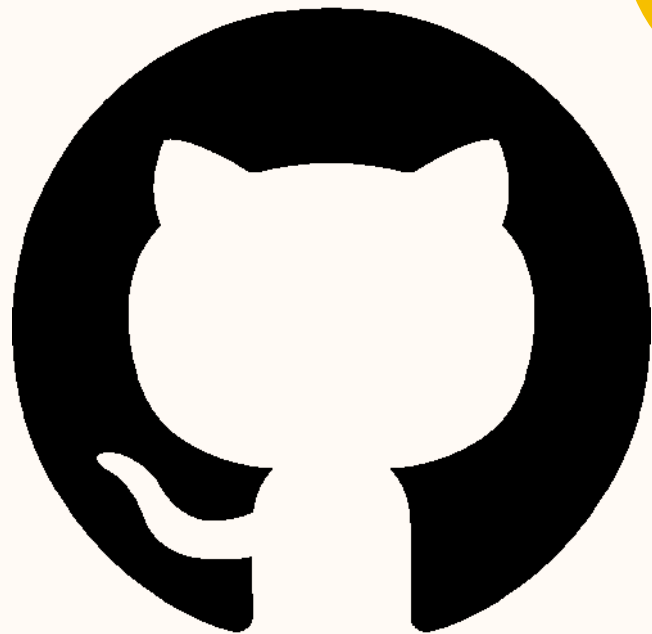
- Config your git.
- Create a new git repository called "ST-project"
- Create a c++ main file that print "Hello world", and use git status to check its status.
- Add main.cpp to your staging area, and use git status to check its status.
- Commit your changes with a suitable message, and use git status to check its status.
- Use git log to display the past commits.



02

Github

Difference between git & github



What is Github

Github is an online platform that allow you to share Git repositories.

Allow you to collaborate with your team and work at the same Git repository online.

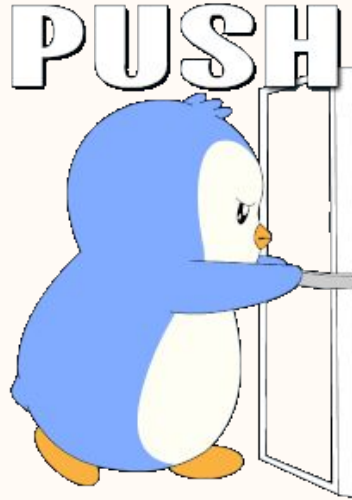
An extra stage is added to publish the changes into the online repository,



So....., What's new?

git clone url

Used to clone a online repository into your device.



git pull **git push**

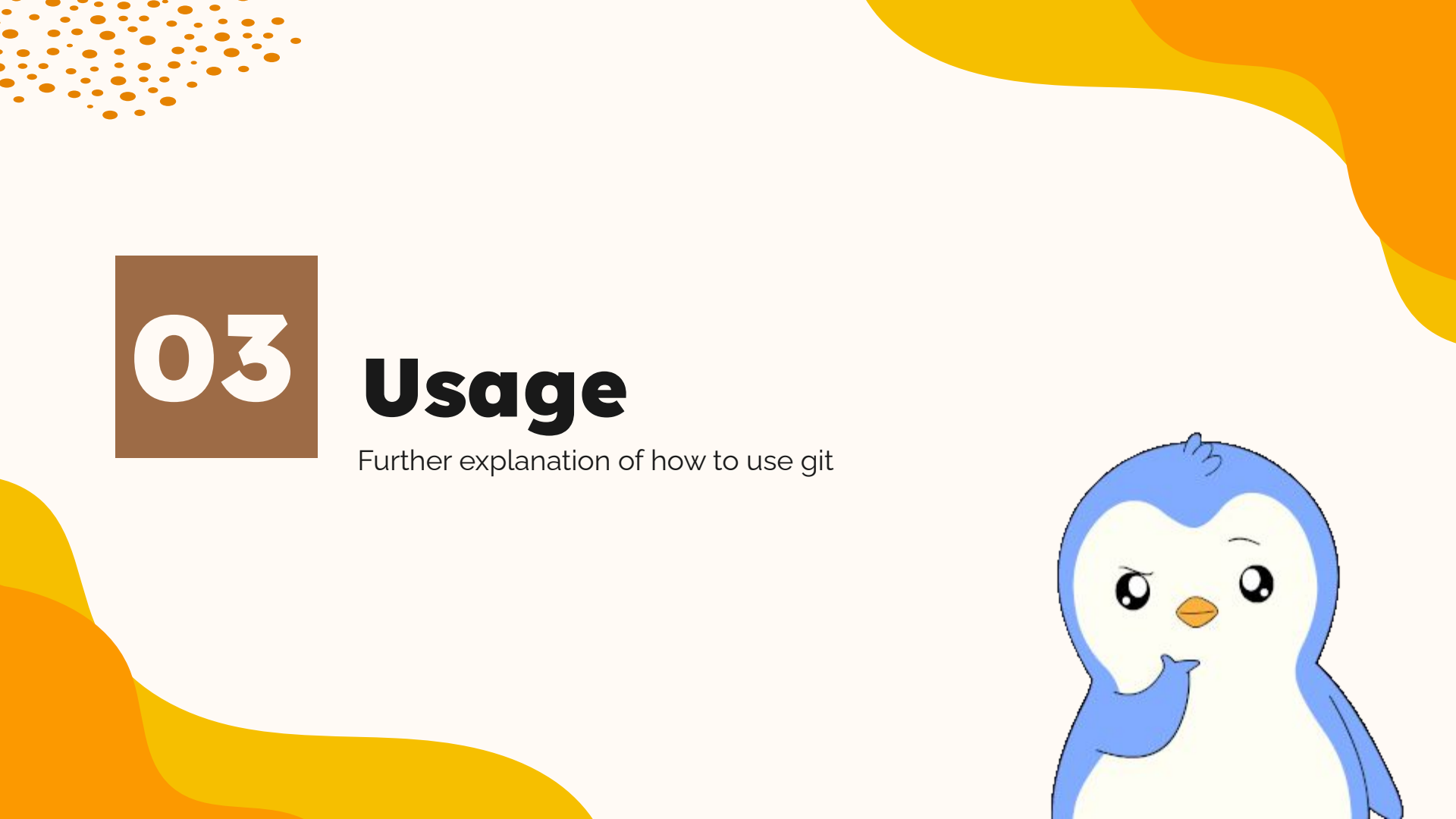
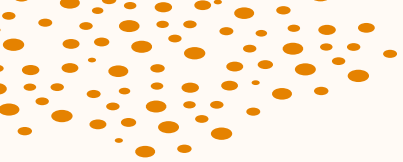
git pull, used to fetch the online repository and download it's new commits into your device.

git push, used to submit your new commits into the online repository.

Hands-on 2



- Create a Github account and login into github.com
- Create a new repository Called "Ibrahim-Yasser" for example (use your name).
- Clone this repository into your device.
- Create a file called "ID.txt" and write your university ID inside.
- Submit your work into the online repository at github and check from the browser.



03

Usage

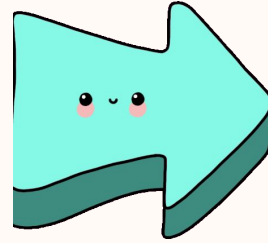
Further explanation of how to use git



What is HEAD?

Head is used to refer for the current commit you are currently working at, usually it's the last commit but you can rollback using the command **git checkout "commit-hash"**

Note: commit-hash can be displayed using **git log**

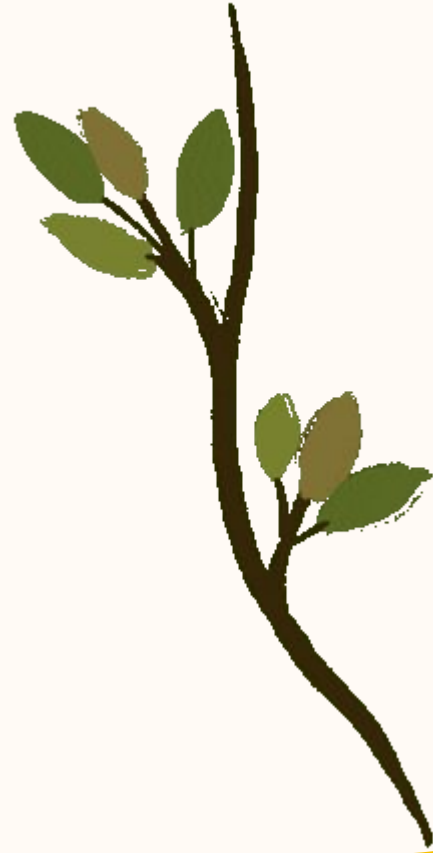




Branching

Suppose you have an idea for a new feature, but you're not sure if it'll work out as you expect, so you do not want to risk modifying your currently working version of the code, This is where branches come in.

To create a new branch, run
git branch <branch-name>





Merge 2 branches

There are 2 ways for merging branches:

- Fast forward (if the new branch is ahead of you)
- 3-way merge (if both branches has new commits)

To merge a branch into your current branch
use:

git branch <branch name>





Hands-on 3

- Use the repository from **Hands-on 1**
 - Create a new branch called **sum** and add a function `sum` to **main.cpp**
 - Return to the **master** branch
 - Create a new branch called **subtract** and add a function **subtract** to **main.cpp**
 - Merge the 2 branches
- 



Thanks!

Do you have any
questions?



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